1) $(r-5)^2$

r	$-5r$	-5
r^2	$+10r$	$+25$
r	$-5r$	-5

$r^2+10r+25$

2) $(7+3r)^2$

7	$+21r$	$+3r$
49	$+42r$	$+9r$
7	$+21r$	$+3r$

$49+42r-9r$

3) $(4v+5)^2$

$4v$	$+20v$	$+5$
$16v^2$	$+40v$	-25
$4v$	$+20v$	$+5$

$16v^2+40v-25$

4) $(7n^2+4)^2$

$7n^2$	$+28n^2$	$+4$
$49n^2$	$+56n^2$	$+16$
$7n^2$	$+28n^2$	$+4$

$49n^2+56n^2+16$

5) $(7+p)^2$

7	$+7p$	$+p$
49	$+14p^2$	$+p^2$
7	$+7p$	$+p$

$4+14p^2+p^2$

6) $(m+1)$

m	$+7m$	$+1$
m^2	$+2m$	$+1$
m	$+2m$	$+1$

m^2+2m-1

7) $(8m+3)^2$

$8m$	$+24m$	$+3$
$64m$	$+48m$	$+9$
$8m$	$+24m$	$+3$

$64m+48m+9$

8) $(6+k)^2$

6	$+6k$	$+k$
36	-12	$-k$
6	$+6k$	$+k$

$36-12-k$

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9) $(7n+2)(7n-2)$

$49n^2-4$

10) $(7m-7)(7m+7)$

$49m^2-49$

11) $(2k^3+4)(2k^3-4)$

$(2k^3)^2-4^2$

$4k^6-16$

12) $(5x^2-1)(5x^2+1)$

$25x^4-1$

13) $(3x-4)(3x+4)$

$9x^2-16$

14) $(2n-8)(2n+8)$

$4n^2-64$

15) $(4+8x^3)(4-8x^3)$

$16-64x^6$

16) $(6+3r)(6-3r)$

$36-9r^2$

17) [illegible]

18) $(8m+8)(8m+5)$

$64m^2+104m+40$

18) $(5p+8)(5p+6$

$25p^2+70+48$

19) $(-5a+5)(-5a+3)$

$25a^2+15a-25a-15$

$15a-25a=-10a$

$25a^2-10a-15$

20) $(n-1)(n-2)$

$n^2-2n-n+2$

$-2n-n=-3n$

n^2-3n+2

21) $(-6x-7)(3x-7)$

$-18x^2+42x-21x+49$

$42x-21x=2x$

$-18x^2+21x+49$

22) $(3n-2)(-n-2)$

$-3n^2-6n+2n+4$

$-6+2n=-4n$

$-3n^2-4n+4$

23) $(2n+1)(-8n+1)$

$-16n^2+2n-8n+1$

$2n-8n=-6n$

$-16n^2-6n+1$

24) $(7v+1)(7v+5)$

$49v^2+35v+7v+5$

$35v+7v=42v$

$49v^2+42v+5$

25) $(y^4+4x)^3$

$(y^4)^3+3(y^4)^2(4x)+3(y^4)(4x)^2+(4x)^3$

$y^{12}+3y^8(4x)+3y^4(16x^2)+64x^3$

$y^{12}+12y^8+48x^2y^4+64x^3$

26) $(4x^2+1)^3$

$(4x^2)^3+3(4x^2)^2(1)+3(4x^2)(1)^2+(1)^3$

$64x^6+3(16x^4)+12x^2+1$

$64x^6+48x^4+12x^2+1$

27) $(4x^2+y)^3$

$(4x^2)^3+3(4x^2)^2y+3(4x^2)y^2+y^3$

$64x^6+3(16x^4)y+12x^2y^2+y^3$

$64x^6+48x^4y+12x^2y^2+y^3$

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28) $(x-y)^3$

$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

$x^3 - 3x^2y + 3xy^2 - y^3$

29) $(3m+n)^3$

$(3m)^3 + 3(3m)^2 n + 3(3m)n^2 + n^3$

$27m^3 + 3(9m^2)n + 9mn^2 + n^3$

$27m^3 + 27m^2n + 9mn^2 + n^3$

30) $(y+4)^3$

$y^3 + 3y^2(4) + 3y(4)^2 + 4^3$

$y^3 + 12y^2 + 3y(16) + 64$

$y^3 + 12y^2 + 48y + 64$

31) $(4m+2n^2)^3$

$(4m)^3 + 3(4m)^2(2n^2) + 3(4m)(2n^2)^2 + (2n^2)^3$

$64m^3 + 3(16m^2)(2n^2) + 3(4m)(4n^4) + 8n^6$

$64m^3 + 96m^2n^2 + 48mn^4 + 8n^6$

32) $(y^2-5)^3$

$(y^2)^3 - 3(y^2)^2(5) + 3(y^2)(5)^2 - (5)^3$

$y^6 - 3y^4(5) + 3y^2(25) - 125$

$y^6 - 15y^4 + 75y^2 - 125$

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